Raymond Opoku

Email • LinkedIn • Website

Experience

College of Medicine - Quality and Patient Safety Initiative, University of Florida Gainesville, FL

Data Scientist IV *July 2026 – Present*

- Data Core Lead: Lead the Data Core of the QPSi Incubator and manage three data scientists. Lead the revamp of the team's data request intake process.
- Security and Compliance Administrator: Serve as QPSi's security and compliance administrator. Provision all Epic build access for the team.
- HiPerGator Administrator: Administer QPSi's allocation on UF's HiPerGator supercomputer, including user access, storage and compute for clinical AI workloads.
- Predictive Model Validation: Validate and monitor Epic Cognitive Computing predictive models in production, and track live alerts across POC, TST and PROD environments.
- AI Governance: Lead the development of AI governance standards, ensuring safety and accountability for machine learning models deployed in clinical settings.
- Host an AI Networking Forum attended by 600+ physicians, administrators and students to foster cross-disciplinary innovation.

Data Scientist III *August 2022 – July 2026*

- Enterprise Strategy: Facilitated high-stakes strategy sessions at the UF Health Readmission Summit, guiding C-suite leadership in defining data-driven interventions for patient access and readmission reduction.
- Epic Solutions Architect: Leveraged certified expertise in Epic Cogito, Caboodle and Clarity to architect automated data pipelines that replaced manual reporting, serving as the technical bridge between raw EHR data and operational dashboards.
- Predictive Analytics Lead: Developed the "Meds Filling" predictive framework, integrating disparate pharmacy data sources (Willow and Surescripts) to flag non-adherent patients for clinical intervention.
- Collaborated with researchers and clinicians to design and implement AI-driven solutions for complex patient safety measures.
- Used advanced SQL and SSMS to extract, clean and validate large-scale clinical datasets for downstream statistical analysis.
- Designed and maintained automated reporting solutions that provided actionable insights for hospital quality improvement initiatives.

Adjunct Professor *August 2022 – Present*

College of Education, University of Florida – Gainesville, FL

- Taught and mentored students in *Design Integrated Media Environments*, fostering innovative media design projects.
- Integrated emerging technology trends into the curriculum to enhance student understanding and creativity.
- Led workshops and seminars, significantly improving student engagement and performance.

Consultant *September 2022*

Gliem Inc. – Gainesville, FL

- Provided consultancy to enhance user experience and learning outcomes by utilizing advanced educational analytics techniques and instructional design principles.

- Collaborated with research teams to evaluate educational strategies and design technology-driven interventions for improving learner engagement.
- Offered recommendations for implementing data-driven decision-making processes to optimize the company's digital platforms.

Data Management Analyst II *June 2020 – August 2022*
UF Online, University of Florida – Gainesville, FL

- Improved data quality through data mining, SQL manipulation and statistical analyses.
- Developed predictive models for student retention, persistence and graduation outcomes.
- Conducted research on student success and federal refugee resettlement program assessments.

Reporting and Data Analyst *January 2020 – June 2020*
Office of Institutional Research, Dakota State University – Madison, SD

- Created dashboards and self-service reports to satisfy institutional data needs.
- Analyzed and interpreted data from multiple sources for diverse stakeholders.

Research and Data Specialist *September 2017 – January 2020*
Office of Institutional Research, South Dakota State University – Brookings, SD

- Designed Tableau dashboards for enrollment, NSSE, and Wokini projects.
- Administered and analyzed institutional surveys to improve student outcomes.

Education

- **Doctor of Philosophy in Educational Technology** – University of Florida (2025)
- **Master of Science in Data Science** – South Dakota State University (2017)
- **Master of Science in Economics** – South Dakota State University (2017)
- **Bachelor of Arts in Economics and Mathematics** – University of Cape Coast (2014)

Certifications and Licences

Epic Cogito | Epic Caboodle | Epic Clarity

Skills

R | Python | Cognos | Qualtrics | STATA | SAS | SQL | Tableau | SPSS

Awards and Achievements

- **Dean's Award for Academic Excellence** – Faculty of Social Sciences, UCC, Ghana (2013, 2014)
- **President** – Economics Graduate Student Association, SDSU (2016-2017)
- **Volunteer Editor** – Wikimedia Ghana

Grants and Fundraising

- National Science Foundation - Innovative Technology Experiences for Students and Teachers (ITEST) ($1,275,109) Personnel Served as support staff for implementation portion of grant via instruction and facilitating workshops. https://education.ufl.edu/sharkai/team/.

Selected Publications

- **Opoku, R.A**, Pei, B., & Xing, W. (2025). **Unveiling Accuracy-Fairness Trade-Offs: Investigating Machine Learning Models in Student Performance Prediction.** *Journal of Learning Analytics. https://doi.org/10.18608/jla.2025.8543*.
- Chen, H., Huang, Y., Yin, C., He, X., Ison, R. L., Bell, L. L., **Opoku, R. A.**, Jiang, Z., Guo, S. J., Liu, M., Tighe, P. J., & Bian, J. (2025). **Care coordination and patient Safety Outcome: a graph-based approach.** *Npj Health Systems, 2(1). https://doi.org/10.1038/s44401-025-00020-9*

- Wusylko, C., Weisberg, L., **Opoku, R.A**, Abramowitz, B., Williams, J., Xing, W., Vu, T., & Vu, M. (2024). **Using machine learning techniques to investigate learner engagement with TikTok media literacy campaigns.** *Journal of Research on Technology in Education, 56*(1), 72–93. https://doi.org/10.1080/15391523.2023.2266518

Selected Presentations

- **Opoku, R.A**, Jones, T., Wusylko, C., Haire, G., Lin, C. H., Adhami, N., King, N., MacFadden, B., Waisome, J., Perez, V., Abramowitz, B., & Killingsworth, S. (2025). **Black Middle Schoolers' Artificial Intelligence Self-Efficacy and Outcome Expectations in a Summer STEM Camp.** Paper to be presented at the *NARST 2025 Annual International Conference*, Washington, D.C.
- **Opoku, R.A**, Jones, T., Wusylko, C., Haire, G., Lin, C. H., Adhami, N., King, N., MacFadden, B., Waisome, J., Perez, V., Abramowitz, B., & Killingsworth, S. (2025). **Black Middle Schoolers' Artificial Intelligence Self-Efficacy and Outcome Expectations in a Summer STEM Camp.** Paper to be presented at the *AERA 2025 Annual International Conference*, Denver, Colorado.
- **Opoku, R.A**, Ma, C., & Moore, R. (2023). **Not THAT Urban Dictionary: Enhancing Online Learning with Dictionary-Based Sentiment.** Presented at the *AECT 2023 Annual International Conference*, Orlando, Florida.
- **Opoku, R.A**, DeLeon, J., & Moore, R. (2023). **Who's Winning: A Review of Student Outcomes in Game-Based Learning.** Presented at the *AECT 2023 Annual International Conference*, Orlando, Florida.
- Moore, R., Miller, C., & **Opoku, R.A** (2022). **A Systematic Review of the Relationship Between Learning Analytics and Course Assessment.** Presented at the *AECT 2022 Annual International Conference*, Las Vegas, Nevada.
- **Opoku, R.** (2022). **Online Computer Science Dropout: A Qualitative Study on Why Online Students Leave the UF Online Computer Science Program.** Presented at the *AECT 2022 Annual International Conference*, Las Vegas, Nevada.

Posters & Abstracts

- **Opoku, R.A** (2023). **Mind the Graph: Using Graph AI to Understand Latent Care Pathways in Surgical Patients at Scale.** Poster presented at the *2023 IHI Annual Conference*, Orlando, Florida.
- **Opoku, R.A**, Williams, R., Nickerson, P., Magoc, T., Tighe, P., Pebe, L., Guo, W., Gupta, S., & Lafitte, C. (2024). **Balancing the Scales of Care: Innovating Fair Predictions for Postoperative Sepsis with AI.** Poster presented at the *2023 AI4Health Annual Conference*, Orlando, Florida.

Invited Talks

- **Opoku, R.A** (October 2026). **Fair AI in Clinical Care: Responsible Use from Principle to Practice.** Lightning talk presented at the *UF College of Medicine AI Day*, Gainesville, FL.
- **Opoku, R.A** (2023). **Let's Build Together: Techniques and Approaches for Engaging Stakeholders in AI Development.** Invited talk presented at the *Empowering Learners for the Age of AI's 2023 Global Online Conference (ELAi)*, Virtual.
- **Opoku, R.A**, Lu, J., & Lazarevic, B. (2022). **Enhancing User Experience and Learning Outcomes: Collaborative Research in Educational Analytics and Instructional Design.** Invited talk presented at *Gliem Inc.*, Gainesville, FL.

Guest Lecture

- **Opoku, R.A** (2025). **Artificial Intelligence (AI) in Medicine: Hope, Hype and Reality: (Frontiers of AI - IDS 1703)** Invited talk presented at *University of Florida*, Virtual.
- **Opoku, R.A**, Jones, T., & Antonenko, P. (May 2025). **Artificial Intelligence (AI) in K-16 STEM Education - EDSC 9870** Invited talk presented at *Georgia State University*, Virtual.